Corrigendum — the appendix has 56 problems, not 54, and problem G2 was never missing

This replaces the earlier addendum. A reader checking my work found that the curriculum appendix contains **56** problems rather than the 54 I reported, and that one of them is problem **B4**, which my verification omitted. Checking further, I found a second and worse error: **problem G2 exists in the appendix**, so the G2 I wrote "because it was missing" was never needed. Both errors came from the same cause and it was mine.

What went wrong

I extracted the appendix's problem list with a pattern that looked for an identifier at the start of a line. Two problems — **B4** and **G2** — begin immediately after a page break, so the character before the identifier is a form feed rather than a newline, and the pattern skipped them. Every count downstream inherited the error: the verification table, the two problem-set introductions, the index, the curriculum's filename, and the claim that G2 was absent.

The original tracker was right all along. Its built-in problem list contains all 56, including B4 and G2 — so the discrepancy between "54" in my documents and "56" in the tracker was visible from the start, and I should have treated the disagreement as a signal rather than assuming my own count was correct.

Correction 1 — problem B4, now verified

B4 · A batch fermentation model from scratch: Monod plus Luedeking–Piret (week 16). Monod growth at μ_{max} = 0.5 h^{-1} and K_S = 0.2 g/L, Luedeking–Piret product formation with α = 0.2 g/g and β = 0.05 $g \cdot g^{-1} \cdot h^{-1}$, $Y_{X/S}$ = 0.45 g/g, from 0.1 g/L of biomass and 20 g/L of substrate.

Quantity	Appendix states	I recomputed	
Time to substrate exhaustion	9.0 h	9.01 h	**matches**
Final biomass	8.0 g/L	8.005 g/L	**matches**
Final titer	2.43 g/L	2.433 g/L	**matches**
Observed biomass yield	0.400 g/g	0.400 g/g	**matches**
Product yield	0.122 g/g	0.122 g/g	**matches**
Volumetric productivity	0.270 $g \cdot L^{-1} \cdot h^{-1}$	0.270 $g \cdot L^{-1} \cdot h^{-1}$	**matches**

So B4 is correct as written, and its teaching point is worth restating because the module repeats it elsewhere: **the observed biomass yield (0.400) is not the yield parameter you specified (0.45)**, because product formation draws its own substrate. Any model with more than one product behaves this way, and a yield reported without saying which products it accounts for is ambiguous. That is the same distinction problem X7 makes between yield on substrate consumed and yield on substrate fed.

Correction 2 — problem G2 was never missing

G2 · Sixteen experiments: factorial grid against Bayesian optimisation is present in the appendix, at its week 41. My earlier addendum stated that G2 was "the only referenced problem missing from the appendix" and supplied a replacement. **That claim was false and the replacement was redundant.**

What to do with the version I wrote: treat it as **an additional problem on the same topic**, not as a substitute. It poses the comparison differently — mine sweeps the acquisition function and the noise level, the appendix's compares a fixed factorial grid with a fixed Bayesian run — so working both is not wasted, but the appendix's version is the one the curriculum intends and the one keyed to week 41.

Correction 3 — the count, and where it was wrong

Document	Said	Now says
Index	54-problem appendix; all 54 verified	**56**
Part 15 (problems X1–X24)	"the 54 in your curriculum's appendix"	**56**
Part 16 (problems Z1–Z9)	"together with the 54"	**56**
Curriculum filename	`…-and-54-problems.pdf`	`…-and-56-problems.pdf`

This verification document	"all 54 problems"	**56**, with B4 added

The verification, restated for all 56

Category	Count	Result
Problems whose numerical answer I could recompute independently	**38**	**All 38 matched**, B4 included
Problems verified in structure only (simulation seeds, qualitative answers, code whose output depends on a random draw)	15	Method and reasoning check out; the printed numbers depend on a seed
Problems with a documented convention difference	3	H2 junction counting, W3 lipid accounting, T1/W1 transport assumptions — all three described in the original addendum and unchanged
Total	**56**	

The three convention differences are not errors in the appendix. H2 counts junctions as n − 1 where I counted n; W3 uses 0.30–0.32 g/g for lipid where my accounting gives 0.275; and T1 and W1 assume different transport parameters from the ones I used. In each case the appendix's convention is internally consistent and is the one to follow when working its problems.

What this says about the rest of the module

An extraction error that silently drops two of fifty-six records is exactly the class of mistake the module's own problems warn about — problem MC1's lesson that a carbon balance passes a reaction that breaks the model, and problem Y2's instruction to run a pipeline on shuffled labels to see whether it leaks. I did not apply that discipline to my own extraction, and the check that would have caught it was available and free: **the tracker's own problem count disagreed with mine, and I did not reconcile them**.

The practical consequence for you is small — no arithmetic anywhere in the module depended on the count, and B4 and G2 are both correct as the appendix wrote them. The reason to record it in full is that a committee will ask how you check your own work, and "I reconcile independent counts of the same thing and investigate disagreements" is a better answer than "I was careful".